A NOETHERIAN ALGEBRAIC SPACE RECEIVING A FINITE SURJECTIVE MORPHISM FROM AN AFFINE SPACE IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Extension from coherent to quasi-coherent modules by filtered colimits, and the affineness criterion by vanishing of H^1. Same chapter spaces-cohomology.tex, labels lemma-directed-colimit-coherent and lemma-colimits (invoked at proof line 3434) and label proposition-vanishing-affine (invoked at proof line 3436). They reduce affineness of X to the vanishing of $H^1(X, \mathcal{F}) = 0$ for all coherent $\mathcal{F}$, which is what the rest of the proof establishes.
- The criterion reducing a property of coherent sheaves on a Noetherian space to irreducible supports. Same chapter spaces-cohomology.tex, label lemma-property-higher-rank-cohomological (invoked at proof lines 3444 and 3453). Conditions (1) and (2) are verified directly for the property $\mathcal{P}(\mathcal{F}) \Leftrightarrow H^1(X, \mathcal{F}) = 0$, and condition (3) is met by the sheaf $\mathcal{G} = f'_*\mathcal{O}_{Z'}$.
- The auxiliary finite morphism over a reduced irreducible closed subspace, and the projection formula for an affine morphism. Same chapter spaces-cohomology.tex, labels lemma-finite-morphism-Noetherian (invoked at proof lines 3450 and 3454) and lemma-affine-morphism-projection-ideal (invoked at proof line 3460). The first supplies Z' with $i' : Z' \to Y$, $f' : Z' \to Z$ and the support property (3)(a); the second identifies $\mathcal{G}' = \mathcal{I}\mathcal{G}$ with $f'_*\mathcal{I}'$.
- Vanishing of the higher direct images of the affine morphisms f' and i, with Leray. Same chapter spaces-cohomology.tex, label lemma-affine-vanishing-higher-direct-images (invoked at proof lines 3470 and 3477) and

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Cohomology on Sites, label sites-cohomology-lemma-apply-Leray (invoked at proof line 3474). Since f' and i are affine, their higher direct images vanish; together with the vanishing of $H^1(Z', \mathcal{I}')$ on the affine space Z' and Leray, this gives $H^1(X, i_*\mathcal{G}') = 0$, which is condition (3)(b).

Source reference. Original lemma 0GF8: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3386–3386. Navigation pointer for \ref{lemma-affine-morphism-projection-ideal} used inside the counted span (line 3460); the argument that consumes it is retained in the span.

Source reference. Original lemma 073H: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1522–1522. Navigation pointer for \ref{lemma-affine-vanishing-higher-direct-images} used inside the counted span (line 3470); the argument that consumes it is retained in the span.

Source reference. Original lemma 073E: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 667–667. Navigation pointer for \ref{lemma-colimits} used inside the counted span (line 3434); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UV: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2866–2866. Navigation pointer for \ref{lemma-directed-colimit-coherent} used inside the counted span (line 3434); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GF7: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3351–3351. Navigation pointer for \ref{lemma-finite-morphism-Noetherian} used inside the counted span (line 3450); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UT: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2652–2652. Navigation pointer for \ref{lemma-property-higher-rank-cohomological} used inside the counted span (line 3444); the argument that consumes it is retained in the span.

Source reference. Original proposition 07V6: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3160–3160. Navigation pointer for \ref{proposition-vanishing-affine} used inside the counted span (line 3436); the argument that consumes it is retained in the span.

Source reference. Original lemma 0733: sources/stacks-chow/tags.txt, source lines 9047–9047. Navigation pointer for \ref{sites-cohomology-lemma-apply-Leray} used inside the counted span (line 3474); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Assume*

(1) *f finite,*
(2) *f surjective,*
(3) *Y affine, and*
(4) *X Noetherian.*

Then X is affine.

Proof. We will prove that under the assumptions of the lemma for any coherent $\mathcal{O}_X$-module $\mathcal{F}$ we have $H^1(X, \mathcal{F}) = 0$. This implies that $H^1(X, \mathcal{F}) = 0$ for every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ by Lemmas 07UV and 073E. Then it follows that X is affine from Proposition 07V6.

Let $\mathcal{P}$ be the property of coherent sheaves $\mathcal{F}$ on X defined by the rule

$$\mathcal{P}(\mathcal{F}) \Leftrightarrow H^1(X, \mathcal{F}) = 0.$$

We are going to apply Lemma 07UT. Thus we have to verify (1), (2) and (3) of that lemma for $\mathcal{P}$. Property (1) follows from the long exact cohomology sequence associated to a short exact sequence of sheaves. Property (2) follows since $H^1(X, -)$ is an additive functor. To see (3) let $i : Z \to X$ be a reduced closed subspace with $|Z|$ irreducible. Let $i' : Z' \to Y$ and $f' : Z' \to Z$ be as in Lemma 0GF7 and set $\mathcal{G} = f'_*\mathcal{O}_{Z'}$. We claim that $\mathcal{G}$ satisfies properties (3)(a) and (3)(b) of Lemma 07UT which will finish the proof. Property (3)(a) we have seen in Lemma 0GF7. To see (3)(b) let $\mathcal{I}$ be a nonzero quasi-coherent sheaf of ideals on Z. Denote $\mathcal{I}' \subset \mathcal{O}_{Z'}$ the quasi-coherent ideal $(f')^{-1}\mathcal{I}\mathcal{O}_{Z'}$, i.e., the image of $(f')^*\mathcal{I} \to \mathcal{O}_{Z'}$. By Lemma 0GF8 we have $f_*\mathcal{I}' = \mathcal{I}\mathcal{G}$. We claim the common value $\mathcal{G}' = \mathcal{I}\mathcal{G} = f'_*\mathcal{I}'$ satisfies the condition expressed in (3)(b). First, it is clear that the support of $\mathcal{G}/\mathcal{G}'$ is contained in the support of $\mathcal{O}_Z/\mathcal{I}$ which is a proper subspace of $|Z|$ as $\mathcal{I}$ is a nonzero ideal sheaf on the reduced and irreducible algebraic space Z. The morphism f' is affine, hence $R^1 f'_*\mathcal{I}' = 0$ by Lemma 073H. As Z' is affine (as a closed subscheme of an affine scheme) we have $H^1(Z', \mathcal{I}') = 0$. Hence the Leray spectral sequence (in the form Cohomology on Sites, Lemma 0733) implies that $H^1(Z, f'_*\mathcal{I}') = 0$. Since $i : Z \to X$ is affine we conclude that $R^1 i_* f'_*\mathcal{I}' = 0$ hence $H^1(X, i_* f'_*\mathcal{I}') = 0$ by Leray again. In other words, we have $H^1(X, i_*\mathcal{G}') = 0$ as desired. □